Research Progress Report

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Contents

Research discussion (w/ Prof. Millán)

- My doctoral goal and first step

Paper search: *Can TMS stimulate memory related areas?*

Experiment follow-up

- Learning EEG recording and *LOOP* software (w/ Satyam, Rawan, Hussein)
- Data processing using *EEGLAB* – preprocessing

Research Discussion: The Grand Goal

Grand goal: Noninvasive brain stimulation for memory dysfunction treatment

Hypothesis

- We can improve memory function (*e.g.* episodic and long-term memory) by stimulating related areas in the brain through closed-loop magnetic stimulation

Grounds

- Memory performance improvement using iEEG-implanted electrodes system (Ezzyat *et al.*, 2017, Ezzyat *et al.*, 2018)
- Lasting enhancement of recollection precision using rTMS (Nilakantan *et al.*, 2018)
- EEG-TMS system enables real time neural stimulation with minimal artifact latency period (10's of *ms*)

Research Discussion: My First Goal

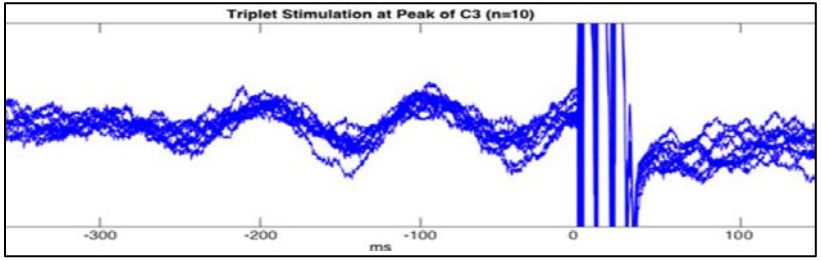
First goal: To modulate the error-related potential (ErrP) using EEG-TMS system

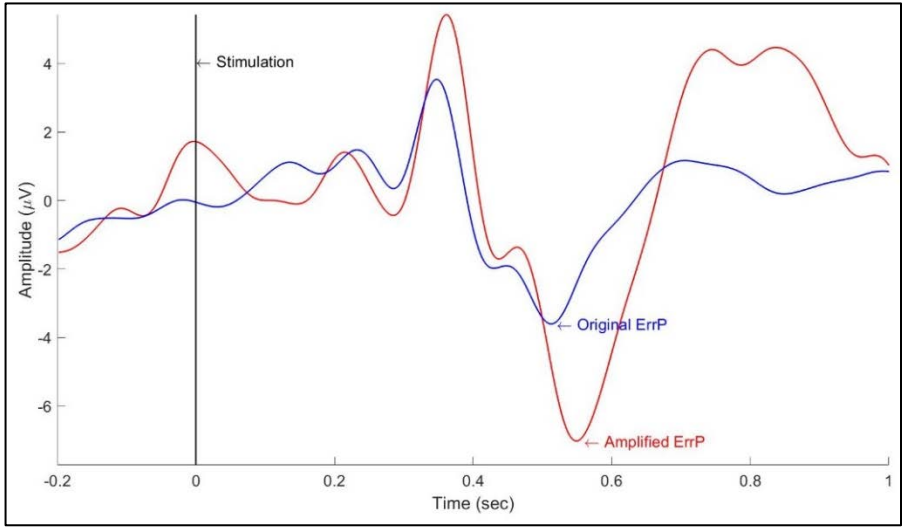
Hypothesis

- EEG-TMS system can detect the ErrP and modify the signal's magnitude and phase for various applications. We can use this system to build a controlled ErrP modulate protocol.

Current limitations

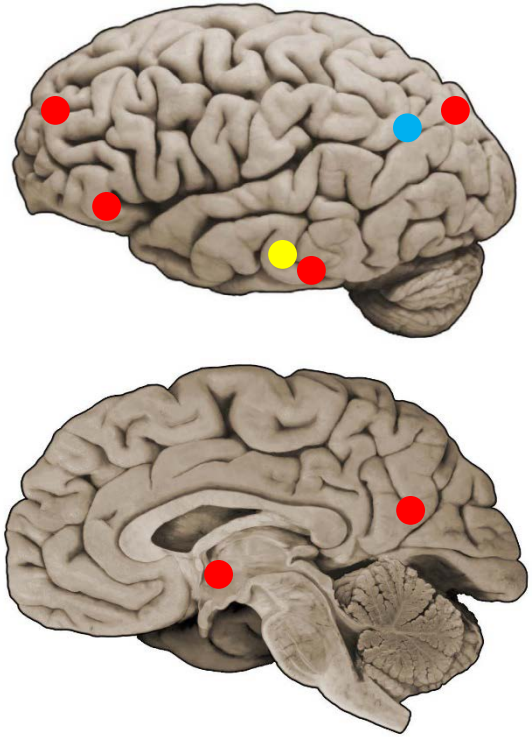
- Inability to record EEG after the stimulation artifact latency period (10's *ms*)





Paper Search: Stimulation for Memory Enhancement

Papers	1. Ezzyat 2017 (●)	2. Ezzyat 2018 (●)	3. Nilakantan 2017 (●)
Recording & Stimulation	iEEG – implanted electrodes		rTMS
Stimulation Region	Hippocampus Entorhinal ctx Perirhinal ctx Parahippocampal ctx Prefrontal ctx Parietal ctx	Lateral temporal cortex	Lateral parietal cortex (Hippocampal posterior-medial network)
Validation Method	Free recall memory task		Object location memory task
Results	Spectral tilt: Increased high-freq. power w/ decrease in low-freq. power	Spectral tilt & Lateral temporal ctx (>HC, MTL)	Evoke potential amplitude and θ-α oscillatory power reduction



- Ezzyat *et al.*, Current biol. 2017
- Ezzyat *et al.*, Nat. Commun. 2018
- Nilakantan *et al.*, Current biol. 2017

Paper Search: TMS Specification

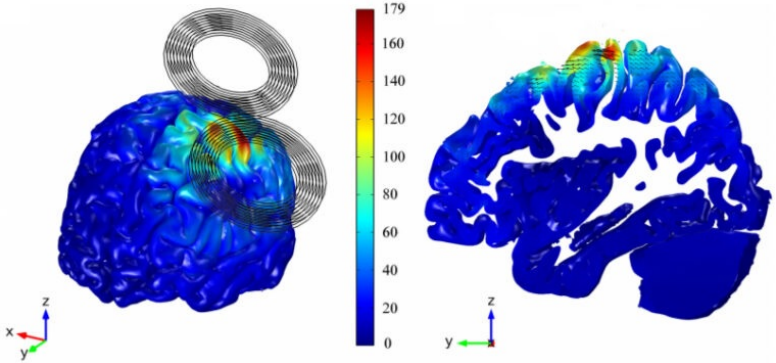
Transcranial magnetic stimulation (TMS)

- Changes the magnetic field which induces a short and highly localized E-field in the cortex
- Interneurons and collaterals of pyramidal cells aligned tangentially to the cortical surface fire

Stimulation depth: 2 – 3 *cm* (Deep TMS: 6 *cm*)

Spatial resolution: 0.5 – 1 *cm* (5T fMRI: 3 – 4 *mm*)

Pulse duration: 5 – 40 *ms*



EEG-triggered TMS system

- Closed-loop magnetic stimulation with EEG recording

Limitations

- TMS saturates the recording amplifier for 10's of *ms* (2016)
- Real-time analysis of data ➔ any filtering will cause phase shifts and edge effects

- Zrenner *et al.*, Front. in Cellular Neurosci. 2016
- Sliwinska *et al.,* J Vis Exp 2014
- Nilakantan *et al.*, Current biol. 2017
- Stefanou et al., J Vis Exp 2019

Plans for Next Steps

Experiment follow-up (w/ Satyam)

- Learn to modify the *LOOP* codes – electrode mask, etc.
- Assist Satyam's MI experiments

Learn TMS operation

Paper search

- Previous studies on applications of rTMS and tES
- Read ErrP-related papers from our group